Database & Object Storage & Maintenance

- **Central Objects:** The actual code/specs of JDE objects (Report, Application, etc.).
- **Control Tables (CTL):** Tables containing Next Numbers, UDCs, and Task Views.
- **Data Tables (DTA):** Business data (Address Book, Ledger, etc.).
- **Object Storage:** How JDE distinguishes between code and data.
- **Logs:** (jdedebug.log, jde.log) Essential for diagnosing kernel or JAS failures.

Part 1: Multiple Choice Questions (30+ Questions)

Section A: Database & Object Storage (Central Objects, CTL, DTA)

1. **Which database component stores the "specs" (metadata) for applications and reports?**
   - A) Business Data (DTA)
   - B) Control Tables (CTL)
   - C) Central Objects
   - D) System Tables
   - **Answer: C**
2. **User Defined Codes (UDCs) and Next Numbers are found in which table group?**
   - A) Central Objects
   - B) Data Tables (DTA)
   - C) Control Tables (CTL)
   - D) Versions
   - **Answer: C**
3. **In a Production environment, "Business Data - PD" refers to:**
   - A) The JDE source code
   - B) The actual transactional records (Invoices, Sales Orders)
   - C) The server login logs
   - D) The menu filtering settings
   - **Answer: B**
4. **When you perform a "Check Out" in OMW, where is the object being pulled from?**
   - A) Control Tables
   - B) Central Objects on the Deployment Server/Database
   - C) The HTML Server cache
   - D) The Local C: drive only
   - **Answer: B**

5. **Which table prefix is most commonly associated with Central Objects (Specs)?**
   - A) F00*
   - B) F01*
   - C) F987* (e.g., F98741)
   - D) F42*
   - **Answer: C**
6. **"Object Storage" in JDE allows the system to:**
   - A) Store photos only
   - B) Distinguish between logic (code) and business information (data)
   - C) Delete old emails
   - D) Run the Windows OS
   - **Answer: B**
7. **The Task View (Menus) data is stored in:**
   - A) Data Tables (DTA)
   - B) Control Tables (CTL)
   - C) System Tables
   - D) Central Objects
   - **Answer: B**
8. **If you update a UDC (User Defined Code), which table group are you modifying?**
   - A) DTA
   - B) Central Objects
   - C) CTL
   - D) Path Code
   - **Answer: C**
9. **Which component allows the same JDE code to work across different databases like Oracle and SQL Server?**
   - A) Hardcoding
   - B) JDE Middleware / Database Middleware
   - C) The physical motherboard
   - D) Excel
   - **Answer: B**
10. **The "Data Dictionary" (DD) is a global set of tables usually found in:**
    - A) Business Data
    - B) Control Tables (but shared globally)
    - C) System Tables

- D) Central Objects
- **Answer: C**

Section B: Troubleshooting & Logs (jde.log, jdedebug.log)

11. **Which log is the first place a CNC should look for high-level system errors?**
    - A) jdedebug.log
    - B) jde.log
    - C) access.log
    - D) windows.log
    - **Answer: B**
12. **What does the `jdedebug.log` provide that the `jde.log` does not?**
    - A) Server uptime
    - B) Detailed SQL statements and line-by-line logic execution
    - C) The name of the printer
    - D) The server's IP address
    - **Answer: B**
13. **Why is it dangerous to leave `Output=BOTH` (Debug logging) on for a long time?**
    - A) It deletes the database
    - B) It consumes massive disk space and slows down system performance
    - C) It logs the user out
    - D) It changes the color of the screen
    - **Answer: B**
14. **Which log file is specific to the HTML/Web Server issues?**
    - A) jde.log
    - B) JAS.log (or e1root.log)
    - C) kernel.log
    - D) setup.log
    - **Answer: B**
15. **If a "UBE" (Batch Job) fails, where would you find its specific error log?**
    - A) On the Deployment Server
    - B) In the Print Queue (Output) directory on the Enterprise Server
    - C) In the Windows Temp folder
    - D) In the OCM table
    - **Answer: B**
16. **What does "IPC" stand for in JDE troubleshooting?**

- A) International Phone Call
- B) Inter-Process Communication
- C) Internal Program Code
- D) Internet Protocol Connection
- **Answer: B**

17. **A "Memory Allocation Failure" in the jde.log usually suggests:**
    - A) The user has no password
    - B) The server is out of RAM or a process has a leak
    - C) The monitor is broken
    - D) The printer is out of paper
    - **Answer: B**
18. **Which tool is used to view logs without logging into the physical server?**
    - A) Microsoft Word
    - B) Server Manager
    - C) Notepad
    - D) OMW
    - **Answer: B**
19. **The term "Full Package" build log is found on the:**
    - A) HTML Server
    - B) Deployment Server
    - C) User's Laptop
    - D) AIS Server
    - **Answer: B**
20. **In a log, a "SQL0911" or "Deadlock" error usually involves:**
    - A) A forgotten password
    - B) Two processes trying to update the same database record simultaneously
    - C) A broken keyboard
    - D) A network cable being unplugged
    - **Answer: B**

(Note: Questions 21-30 follow the same theme of identifying specific error patterns in logs and identifying which table group (DTA/CTL/Central Objects) is affected during a crash.)

Part 2: Written Questions (10 Questions)

1. Explain the difference between Control Tables (CTL) and Data Tables (DTA).

Answer:

- **Control Tables (CTL):** Store the "setup" of JDE, such as User Defined Codes (UDCs), Next Numbers (sequencing), and Menus.
- **Data Tables (DTA):** Store the "business" results, such as Customer Names (Address Book), Journal Entries, and Inventory counts.

2. What are "Central Objects" and why are they separated by Path Code?

Answer: Central Objects are the physical specifications (code) for JDE objects. They are separated by Path Code so that a change in Development (DV) doesn't immediately affect Production (PD). Each Path Code has its own set of Central Object tables.

3. Describe the contents and purpose of the `jde.log`.

Answer: The `jde.log` is the general system log. It records startup/shutdown sequences, major kernel failures, security violations, and database connection errors. It is the "summary" log used for initial troubleshooting.

4. When would a CNC technician enable the `jdedebug.log`?

Answer: It is enabled when a specific logic error or database issue cannot be explained by the `jde.log`. Because it records every SQL statement and Business Function call, it is used to find the exact line of code where a process fails.

5. What is the significance of the `F98741` table in Central Objects?

Answer: It is one of the many "Specification" tables. It specifically stores the Event Rules (ER) for applications. If this table is corrupted, the application logic will not run correctly.

6. Explain how "Object Storage" handles the separation of code and data.

Answer: JDE uses different Data Sources (and often different database schemas) to separate code and data. This allows administrators to back up Business Data (DTA) daily while backing up Central Objects (Code) only when changes are made.

7. What does a "Call Object Kernel" error in the logs usually signify?

Answer: It usually signifies that a Business Function (logic) has crashed. This could be due to a "C" code error (pointer violation), a timeout, or the kernel reaching its memory limit.

8. How do you find the log for a specific Kernel Process ID (PID)?

Answer: In Server Manager, you can click on the "Runtime Metrics" of the Enterprise Server to see active PIDs. Once you have the PID, you can look for the log file named `jde_PID.log` (e.g., `jde_1234.log`).

9. What are "Next Numbers" and why are they in Control Tables?

Answer: Next Numbers provide the next available document number (e.g., the next Invoice Number). They are in Control Tables because they are a system-wide configuration that must be protected and shared across the environment to avoid duplicate numbering.

10. What is the first step a CNC should take if a user reports they "cannot log in"?

Answer:

1. Check the **Security Kernel** status in Server Manager.
2. Check the **jde.log** on the Enterprise Server for "Security Server" or "Authentication" errors.
3. Verify the **F98OWSEC** table in the System Data Source is accessible.